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Editors-in-Chief,
Journal of Functional Analysis
Prof. Alexandru D. Ionescu, Prof. Natasa Sesum, and Prof. Stefaan Vaes

Dear Editors-in-Chief of the Journal of Functional Analysis,

I am writing to submit the manuscript "Shorted Schur Complements for Trace-Class Covariance Blocks and Singular Gaussian Conditioning" as an original research article for JFA.

The manuscript is about positive block operators and inverse-free Schur complementation. In finite dimensions, the Schur complement is written as

$$A - BD^{-1}B^*.$$

For compact trace-class Hilbert covariance blocks, this expression is not a valid ambient operator formula: the complementary block has no bounded inverse on the full state space. The inverse-free Schur object is then the classical Krein–Anderson–Trapp shorted operator on the Cameron–Martin range; taking this framework as background, the manuscript determines the sharp quantitative behaviour of the short along the singular path and its Gaussian finite-part consequence.

Our results provide:

1. a rigorous shorting reformulation of the block-triangular OU regularization and rank-one block-operator setup;
2. the affine finite-part scaling law $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$ and the sharp singular-path constant of the shorted residual,
$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \qquad \lambda^6 R_\lambda \to \frac{\varepsilon_0^2}{4}\,|a_{Ik}|^2_{Q_0},$$
obtained from a two-sided Cameron–Martin energy bound;
3. a Galerkin consistency theory via nested Rayleigh–Ritz truncations, avoiding direct matrix inversion;
4. downstream Gaussian finite-part asymptotics for rank-one disintegration, namely the exact identity $\mathfrak{J}_\lambda = \frac{b_k^2}{\varepsilon_0} - \frac{1}{2}\log(1-\rho_\lambda)$ and the sharp correction
$$\lambda^4\left(\mathfrak{J}_\lambda(\mu_{\mathrm{obs}};\mu_\lambda) - \frac{b_k^2}{\varepsilon_0}\right) \to \frac{\varepsilon_0}{4}\,|a_{Ik}|^2_{Q_0},$$
obtained from disintegration and Fredholm determinant/trace cancellations.

Gaussian conditioning enters as the main Radon-law consequence of this operator theorem. Disintegration supplies the exact-evidence measure, while Fredholm determinant/trace cancellations identify the finite part. We also include a concrete 1D elliptic Gaussian field example to demonstrate the framework in a standard trace-class/SPDE setting.

The manuscript is closely aligned with JFA's scope through its focus on positive block operators, shorted operators, Cameron–Martin geometry, trace-class covariance, and Hilbert-space variational methods; the probabilistic application is a major consequence of the operator-theoretic core.

Thank you for your consideration.

Sincerely,
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